Foundations of Data Engineering and Analytics

Data Analysis Project: Datasheet for the CT7 Energy Windows Dataset

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This datasheet describes the dataset `ct7_windows_5s.csv`, the raw excerpt it is derived from and the intermediate files. It follows the questions of Gebru et al. [1].

Motivation

Purpose. The dataset was created for a course project that asks how well the net battery energy of a Formula Student electric car over the next 5 seconds of driving can be predicted from the preceding 5 seconds. No existing dataset contains such data for this car.

Creator and funding. Created by the project group (Lucas Alois Griesser, Luca Jenewein, Nikola Adžić) from the telemetry logs of the 2025 car CT7 of the Formula Student team CTM, to which Lucas Alois Griesser belongs. There was no funding.

Composition

Instances. Each row is a pair of consecutive, non-overlapping 5-second windows from one driving stint: features describe window w, the target is the net battery energy of window $w{+}1$. The dataset has 4,401 rows and 16 columns (0.36 MiB) from 315 stints on 11 test days, about 6.8 hours of driving under power. The intermediate file `ct7_clean_10hz.csv.gz` contains the cleaned 10 Hz time series of the same stints for plots of whole drives.

Files. The submission contains the raw excerpt `data/raw/S###.log.xz` (5,816,533 original `candump` lines from 120 log files, 36.5 MiB compressed), the definitions of the 10 CAN messages used (`data/ct7_signals.dbc`, taken from CTM's DBC file with unchanged IDs, bit layouts and scaling), the counts of the full scan (`ct7_scan_stats.csv`, `ct7_archive_summary.csv`), the dataset and the cleaned 10 Hz file. `00_extraction.ipynb` decodes the excerpt into 7,041,760 values (`ct7_can_extract.csv.gz`) and a session table (`ct7_sessions.csv`); these two files are not included, because the notebook recreates them. `01_preprocessing.ipynb` turns them into the dataset.

Fields.

- Identifiers: `day_id` (test day, D01, D02, … in date order; the dataset contains D03, D05, D10 and D13 to D20), `session_id` (one power-up of the logger), `stint_id` (one continuous drive), `window` (index within the stint).
- Features of window w: mean motor speed and its change (`rpm_mean`, `rpm_delta`, 1/min); accelerator pedal mean and maximum (`accel_mean`, `accel_max`), brake pedal mean (`brake_mean`) and regeneration request mean (`regen_mean`), all in %; mean absolute steering angle (`steer_abs_mean`, °); battery voltage at the start (`voltage_start`, V); highest cell temperature (`cell_temp_max`, °C); net battery energy (`net_energy_kj`, kJ); time since the start of the stint (`t_in_stint_s`, s).
- Target: `net_energy_next_kj`, the net battery energy of window $w{+}1$ in kJ: $E = \sum_k P_k \, \Delta t$ with $\Delta t = 0.1$ s and P the battery power measured by the ISA sensor. $E > 0$: the battery was discharged; $E < 0$: regenerative braking fed back more energy than the car used.

Sample or complete set. The dataset covers all logged driving under power from 26 May to 16 August 2025 that passes the filters described below. A wheel motor turned on 20 days; 9 of them contain no usable driving under power (only the front motors logged on one day, otherwise less than a minute above 300 1/min or no drive power), so 11 days remain. The dataset is not

representative of competition driving: all sessions are test drives, and the days differ strongly in the amount of driving.

Missing information. The steering angle is missing until 21 June 2025, because the CAN message had an older layout without it. Ambient temperature, position on track, lap times, driver, track layout and car settings were not logged or are not part of the dataset.

Errors and noise. The motor speed is used as a speed proxy and includes wheel slip. Sentinel values, frozen motor speeds and corrupted frames were set to missing, and pairs with missing or inconsistent power values were removed (see Preprocessing).

Dependencies. Consecutive rows of one stint are not independent (speed, battery voltage and temperature change little from one window to the next), and rows from the same day share track, driver and settings. Training and test data should therefore be split by `day_id`, not by row.

Confidentiality and personal data. The data are internal engineering data of CTM, shared for this course with CTM's permission. They contain no names and no positions, and the driver is not recorded. The derived files contain no absolute timestamps, but the raw excerpt keeps the original Unix timestamps of the frames, and `ct7_scan_stats.csv` lists the file date of each session, so the dates and times of the test drives can be recovered. The data are therefore pseudonymous rather than anonymous: someone with CTM's test schedule could link a day, and possibly its driver, to the rows.

Collection process

Acquisition. A Raspberry Pi in the car recorded every frame of the four CAN buses (`can-main`, `can-inv`, `can-sense`, `can-ams`) with `candump`, one log file per power-up. The signals come from the four wheel hub motor inverters, the pedal box, the front vehicle control unit (steering angle), the ISA current and voltage sensor of the battery and the battery management system.

Time frame and sampling. The logs are named 24 April to 16 August 2025 (528 files, about 31 GiB; the files before 7 May are empty). The dataset uses the 342 non-empty files named 26 May 2025 or later (27.1 GiB). The 130 earlier non-empty files contain no driving with a battery power measurement: ISA frames appear only in 26 files from 24 and 25 May, and in the four of them in which a wheel motor turns, a single motor exceeds 100 1/min for at most 4 s (up to 666 1/min) while the battery current stays below 0.1 A. This was checked once on the full logs and cannot be repeated with the package. The logging computer has no battery-backed clock, so its time can jump forward after booting, and a file name can show an earlier date than the data. The extraction therefore takes the date of a session from its first active timestamp, and the pre-processing starts a new recording block at every gap over 2 s.

Activity filter. The scan kept at most one frame per message and 100 ms, and only periods in which a wheel motor turned faster than 100 1/min, plus 30 s before and after (35.5 % of the thinned frames of the 120 logs with activity). Every stint row lies within 21 s of such a motor speed, so the filter removes no driving. While the car stands within the kept periods (8.5 h), the mean battery power is close to zero, so the standing time that was not kept very likely holds little energy as well; this is an estimate, because the power outside the kept periods was not extracted.

Who collected the data. The data were recorded automatically by the team's logging system during test drives by team members. No separate ethical review was carried out; the dataset contains no direct personal identifiers (see Confidentiality above).

Preprocessing

Two notebooks document every step, and both run from the submitted files. The raw excerpt, the message definitions and the scan counts were written once from CTM's full logs and DBC file by `make_raw_excerpt.py`, which is included to document their origin.

1. `00_extraction.ipynb`: decoding of the raw excerpt into 12 signals with the message definitions in `ct7_signals.dbc`, the same 100 ms thinning and activity rule as the scan (applying them again changes nothing), timestamps relative to the earliest thinned frame of the 10 messages in each log file, and pseudonymous session and day IDs. Output: `ct7_can_extract.csv.gz`, `ct7_sessions.csv`.
2. `01_preprocessing.ipynb`: checks of the timestamps (no backward steps within a signal, no duplicates) and recording blocks at gaps over 2 s, so that values before and after a clock jump are never combined. Fixed bounds and sentinel values: 646 steering angles, 4 power values, 1 current value and 2,808 cell temperatures of 0.000 °C are set to missing, and so are 1,083 frozen motor speed values. Alignment on a 10 Hz grid, using no value older than 1 s, and a consistency check of the power against voltage times current. Stints: mean motor speed of at least three motors above 300 1/min, stops shorter than 20 s, at least 10 s long, and a peak battery power above 1 kW (30 of 345 stints fail this). 5-second windows; pairs with incomplete power or missing required values (13 pairs) or inconsistent energy (28 pairs) are removed. The target and three features are recomputed for 20 random rows as a leakage check. Missing values are not imputed. The only missing values left are in `steer_abs_mean` (847 rows: 787 from days before 21 June 2025 and 60 from that day).

Raw data. CTM keeps the full logs (31 GiB); the excerpt keeps every frame the extraction uses in its original form.

Uses

Intended use. Short-term forecasting of the net battery energy and the analysis of its relation to driver inputs and vehicle state within this course project.

Limitations of use. Results do not transfer to other cars, to competition conditions or to other seasons without new data. The dataset must not be used for decisions that affect the safety of the car.

Distribution

The dataset is not public. It is submitted only to the course instructors via OLAT, with CTM's permission, together with the raw excerpt and the definitions of the 10 CAN messages used. The full logs and CTM's complete DBC file are not part of the submission.

Maintenance

The dataset is maintained by the project group for the duration of the course (until January 2027). It is regenerated by rerunning the two notebooks on the included excerpt; only a new excerpt needs CTM's full logs. Changes are documented in the notebooks.

References

[1] T. Gebru, J. Morgenstern, B. Vecchione, J. Wortman Vaughan, H. Wallach, H. Daumé III, K. Crawford. Datasheets for datasets. arXiv:1803.09010v8, 2021. `https://arxiv.org/abs/1803.09010`